Earth-State Embeddings:
self-supervised representations of the observed ocean, probed against the Atlantic overturning circulation

blauewelt/earth — technical report, v4

8 August 2026 (evening revision)

Abstract

We train a small self-supervised model to compress everything the observing system knows about each point of the ocean, each month, into a single vector — an *embedding* — and then ask whether those vectors, which never see any transport measurement, carry the Atlantic Meridional Overturning Circulation (AMOC). They do: a linear read-out from the embeddings along 26.5°N attains a year-blocked cross-validated correlation of $r = 0.60$ with the monthly, deseasonalized RAPID transport series — to our knowledge the first cross-validated monthly-scale figure against real RAPID data from inputs that never include RAPID. A temporal model over the frozen embeddings predicts the ocean's anomaly field beyond a year ahead (anomaly correlation 0.62 at one month, crossing the conventional 0.5 useful-skill line near four months), while the transport read-out loses skill after about one season, consistent with its wind-driven monthly variance. This revision adds a *probe ladder* — ridge, small MLP, and a supervised attention head over the *unpooled* section — and it rewrites the story of the week. Pointwise nonlinearity recovers nothing anywhere: the MLP trails the ridge on every target, for every codec. But the attention head, which sees the ∼67 section pixels individually instead of their mean, raises every codec — and the codec whose channel tokens carry 3×3 spatial patches reaches $\mathbf{r = 0.690}$ [0.570, 0.781], RMSE 2.02 Sv against a 2.79 Sv target, the best transport figure of the programme. The mechanism is thermal wind: geostrophic transport is the east-minus-west density difference across the section, and mean-pooling computes exactly the statistic in which that difference cancels. An end-to-end control then decomposes the headline: the same head over *raw* section data reaches 0.659 at matched receptive field, so pretraining's own margin on this task is +0.031 and not significant — the codec's demonstrated value lies in field prediction and transfer, not in the monthly RAPID probe. An apparent finding that expanding the input from 14 to 24–25 channels had *cost* monthly RAPID skill is retracted as a representation claim — the pooled probes were billing relocated information as lost — while the genuine gain those channels brought at the deep MOVE array at 16°N ($r = 0.206$ [0.044, 0.346], our first multi-target confidence interval excluding zero) stands.

1 Introduction

The AMOC moves on the order of 17 Sv of water northward through the Atlantic's upper kilometre ($1\,\mathrm{Sv} = 10^6\,\mathrm{m}^3\,\mathrm{s}^{-1}$), carries roughly a petawatt of heat, and is directly measured at only a handful of latitudes, for only about two decades. It is the canonical example of a climate-critical quantity that is *composite*: part wind-driven (Ekman transport), part density-driven (geostrophic shear), part boundary currents. No single satellite or float sees it; it must be inferred from the state of the ocean around it.

This project asks a representation-learning question of that inference problem: *if we compress the whole observable ocean state into generic, task-blind embeddings, do climate-critical integrals like the AMOC come along for free?* The design is deliberately strict: the embedding model never sees any transport data; every read-out is a linear (ridge) regression, so any skill must already be present, linearly accessible, in the representation; and every claim is cross-validated in blocks that respect the autocorrelation of the climate system. The project also serves as a live test of a foundation-model hypothesis (§6): that a single representation trained on *all* the data — every ocean, every channel — serves specific regional tasks at least as well as one trained only on the task's own region.

2 Terminology

Terms are used throughout in the following precise senses.

Transport the physical quantity all "truth" targets measure: the volume of water a current system moves, in Sverdrups ($1\,\mathrm{Sv} = 10^6\,\mathrm{m}^3\,\mathrm{s}^{-1}$). The AMOC's strength is a transport ($\sim$17 Sv northward across 26.5°N). A *transport series* is one array's monthly record of it; none is ever a model input.

Fold one round of the year-blocked cross-validation: a calendar year serves as the test set while the probe is re-fitted from scratch on all other years. "Per fold" means exactly that re-fitting — no test month ever influences the weights that predict it.

Transport claims / field-prediction claims the project's two scoreboards, kept apart because their baselines differ. Transport claims: predicting the mooring series (k-fold r, RMSE in Sv, event capture) against the wind-only ridge. Field-prediction claims: predicting the ocean's own input fields ahead in time (chan%, rollout MSSS/ACC) against persistence and climatology.

Channel one physical input field: e.g. surface current speed, mixed-layer depth, temperature at 700 dbar, zonal wind stress. The data tensor has 14 channels in the first generation and 24–25 in the second (Table 1, §3.1).

Pixel one $1° \times 1°$ ocean grid cell. A *pixel-month* is one pixel at one month — the atomic sample.

Channel space the space of the physical fields themselves. An error "in channel space" is an error in predicted physical values (normalized units), directly comparable with a persistence forecast of the same fields.

Embedding / embedding space the 64-dimensional vector z the encoder produces for a pixel-month, and the space of such vectors. An error "in embedding space" compares predicted to actual z; it is an internal diagnostic, since z has no physical units.

Codec the stage-1 encoder–decoder (§4); it encodes a pixel-month's channels to z and can decode z back to channels. "Codec" emphasizes that it is a learned compression.

Anomaly space all dynamic channels are expressed as departures from that pixel's own monthly climatology (computed over training years only), then normalized. This removes everything predictable from location and calendar alone.

Climatology the long-term mean for a given pixel and calendar month. As a forecast, "climatology" predicts zero anomaly — the no-skill floor every real forecast must beat.

Persistence the forecast that the anomaly stays exactly as it is now. Hard to beat at short range; decays into a poor forecast at long range. *Damped persistence* multiplies the anomaly by its own lag-h autocorrelation — the fair statistical baseline.

Patch codec a codec variant whose channel token carries the 3×3 neighbourhood of the pixel (nine values and nine observed-flags) rather than a single scalar, giving the encoder gradient vision and turning the representation from an expander into a genuine compressor (§4).

Stage 2 / temporal model a causal transformer over a pixel's sequence of frozen embeddings that predicts next month's embedding. "Frozen" means the codec's weights are not updated by stage-2 training, so improvements are attributable.

Probe a read-out fitted from frozen embeddings to a target series; the codec and stage 2 are never updated by it. Historically a ridge regression, deliberately weak; now the bottom rung of the probe ladder below. A probe measures what a representation *contains*.

Probe ladder three read-outs of the same frozen embeddings, increasing only in what they may see: *ridge* (linear, on the section-mean embedding), *MLP* (adds pointwise nonlinearity, same pooled input), *attention head* (adds spatial structure: one learned query attends over the unpooled per-pixel embeddings of the section). Each rung localizes where a capability comes from.

Section the row of grid cells nearest a mooring array's latitude, clipped to the array's longitude span (e.g. RAPID: 26.5°N, 80°W–13°W). Probes read the mean embedding along the target's section.

channel(s)	source	record	role
surface current speed, MLD	GLORYS reanalysis	1993–	dynamics
T at 4 or 8 pressure levels	Roemmich–Gilson Argo	2004–	density structure
S at 4 or 8 pressure levels	Roemmich–Gilson Argo	2004–	density structure
wind stress τ_x, τ_y	NCEP R1	1948–	Ekman forcing
within-month τ std. dev.	NCEP R1 daily	1948–	storminess
monthly SST (optional)	OISST v2.1	1981–	thermal memory
SST, precip climatologies	OISST, GPCP	fixed	(measured inert; §8)

Table 1: Channels of the data tensor: monthly, 1°, 1993–present. The pilot window was the North Atlantic (100°W–20°E, 0–70°N; 5,787 ocean pixels); the default window is now global (44,964 pixels). Missing data (Argo before 2004, polar gaps) is represented explicitly, not imputed. The Argo level count changed from 4 to 8 pressures partway through the programme, which is what separates the two channel families of §3.1.

Holdout years entire calendar years (2009, 2017, 2023) excluded from all training; with a held-out mid-Atlantic longitude band they form the blocked validation splits. Random splits are never used — adjacent months are too correlated to count as independent.

k-fold (year-blocked) every calendar year takes one turn as the test set while the probe is fit on the others; the assembled out-of-fold predictions are scored once. Confidence intervals come from a bootstrap that resamples whole years (a *block bootstrap*).

chan% held-out next-month prediction error reduction versus persistence, in channel space, in anomaly space: $100\,(1-\mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{pers}})$. Comparable only within one channel set, since the persistence baseline moves when channels change.

MSSS mean squared skill score, $1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{ref}}$ [6]; our rollout reports it against climatology, persistence, and damped persistence.

ACC anomaly correlation coefficient: the Pearson correlation between predicted and observed anomalies at a given forecast horizon. ACC = 0.5 is the conventional "useful skill" line.

RAPID / FC / MOVE / OSNAP / SAMBA the five transport truth series: the RAPID array at 26.5°N (2004–, monthly), the Florida Current cable at 27°N (1982–, daily), MOVE at 16°N (2000–2022, the *deep* western-boundary flow), OSNAP in the subpolar gyre (2014–2022), SAMBA at 34.5°S (2009–2017). None is ever an input.

3 Data

The tensor (Table 1) is built entirely from credential-free public archives. Two properties matter most. First, *missingness is information*: a pixel-month's unobserved channels enter the encoder as learned "missing" tokens rather than imputed values, so the representation knows what the observing system knew. Second, everything is in anomaly space (§2) — the founding negative result of the project was that a model trained on raw fields wins reconstruction by memorizing the seasonal cycle while learning nothing about change.

The five truth series (RAPID, FC, MOVE, OSNAP, SAMBA) ride along as never-input targets: 240 RAPID months plus 792 additional truth months across four other latitudes.

3.1 Two channel families, and why they may not be compared

The tensor is rebuilt from the fetcher scripts at the start of every training job, so a change to a fetcher silently changes the input of every subsequent run. Sampling 8 Argo pressure levels instead of 4 took the tensor from 14 channels to 24 (25 with monthly SST) without any dispatch asking for it. Every result below therefore belongs to one of two families, and cross-family comparison is not licensed: the persistence baseline that chan% is measured against moves with the channel set, and, as §7.4 shows, so does the meaning of the transport probe. Channel counts in this report are read from each checkpoint, never from a run's name — a run directory carried a misleading name for an hour before this was enforced.

4 Methods

Stage 1 — the codec. A per-pixel masked autoencoder producing $z \in \mathbb{R}^{64}$ per pixel-month, trained on all pixel-months outside the holdout blocks. The pilot is 0.92M parameters; a 10.25M variant (320×8, the size the Chinchilla-style arithmetic of the Compute paragraph anchors for the current tensor) is training at time of writing. Checkpoints carry their architecture, and every loader reconstructs from it.

What exactly enters the encoder. The encoder is a small transformer (128×4 in the pilot, 320×8 in the 10M variant) over $C{+}2$ tokens for one pixel-month:

- **One token per channel**. If the channel is observed and not masked, the token is a linear projection of its scalar value (in anomaly space: the departure from that pixel's own monthly climatology, standardized) plus a learned per-channel identity embedding. If the channel is *masked for training*, the value is replaced by a learned `mask` token; if it is *genuinely unobserved* — Argo before 2004, polar gaps — by a distinct learned `missing` token. "I am hiding this from you" and "the observing system never measured this" are different facts, and the distinction measurably matters.
- **One context token**: a linear projection of $[\sin(2\pi m/12),\ \cos(2\pi m/12),\ \mathrm{lat}/90,\ \mathrm{lon}/180]$ — season and position. Never masked. This token is why static climatology input channels carry zero information (§8).
- **One CLS token**, whose final hidden state is projected to $z \in \mathbb{R}^{64}$.

The decoder never sees the inputs — it answers *queries* (channel, Δlon, Δlat, Δmonth) from z alone, with offsets up to ± 3 cells and months, so the training signal is reconstruction of masked channels *plus* prediction of spatial and temporal neighbours.

The patch codec. At 14 channels a pixel-month carries on average 10.4 observed values, so a 64-dimensional z is an *expander* ($\sim 6\times$), as in masked language models; the information bottleneck is the masking objective, not the dimensionality. The patch variant gives each channel token its 3×3 neighbourhood — nine values and nine observed-flags, projected by $\mathrm{Linear}(2p^2, d_{\mathrm{model}})$ — so that at 24 channels roughly 160 observed values enter one 64-dimensional vector, a genuine 2.5:1 compression. Longitude wraps (the globe is periodic in x) and latitude clamps with out-of-range rows marked unobserved; the centre cell defines whether a channel counts as observed, and reconstruction targets remain centre values, so the loss is unchanged and $p{=}1$ checkpoints load identically. The motivation is physical as well as informational: thermal wind *is* a density gradient, and a gradient does not exist within a single pixel.

Stage 2 — the temporal model. A causal transformer (default $d_{\mathrm{model}}{=}96 \times 3$ layers, 0.35M parameters; capacity sweep in Fig. 6) over each pixel's 24-month embedding sequence, predicting z_{t+1}; scored after decoding to channel space. The codec stays frozen.

Probes — the ladder. The base instrument is a ridge regression from section-mean embeddings to each truth series, deseasonalized, regularization chosen on a training tail — never on test data; verification is year-blocked k-fold with block-bootstrap confidence intervals. Two stronger read-outs sit above it, behind the same folds and the same inner-tail discipline. A small MLP ($64 \to 32 \to 1$, weight decay, early stopping, three seeds per fold) tests whether skill is present but not *linearly* accessible. An attention head tests whether skill is present but not accessible from the *pooled* section: every (pixel, month) embedding enters as a token with a longitude encoding, one learned query cross-attends over them, and a two-layer read-out maps the pooled vector to transport ($\sim$25k parameters, weight decay 10^{-2}, token dropout, three seeds per fold). The codec stays frozen throughout — these are probes, not fine-tuning. The design principle: each rung adds one capability, so a gap between rungs is attributable to that capability alone. Beneath the ladder sit two end-to-end *controls*: the identical head fed raw section values and observed-flags instead of embeddings, at single-pixel and at 3×3 receptive fields — pretraining is then the only manipulated variable (§7.3).

Rollout. For horizons $h = 1 \dots 12$: initialize with true embeddings, predict one month, feed the prediction back, repeat — scored in channel space against climatology, persistence, and damped persistence, and probed for the AMOC at each horizon.

Compute. Runs through 2026-08-07 were trained on GitHub-hosted 4-core CPU runners at ∼1.3 steps/s. Since then three rented consumer GPUs (RTX 4090, ∼\$0.24/h each) are attached as self-hosted runners at ∼124 steps/s each — a factor of 95 per box, which turns a 30k-step schedule from six hours into four minutes and made the controlled comparisons of §7 affordable. The raw source data (∼13 GB) is mirrored as a GitHub release so a fresh box seeds its cache from a CDN rather than the origin archives, one of which (the Argo host) times out under load. Nothing about the model or protocol changed with the hardware. Chinchilla-style anchor for sizing: the current tensor carries ∼270M observed values per epoch; at the ∼20:1 data-to-params ratio that suggests a ∼13M-parameter codec — 14× the pilot, and the reason the 10M variant exists. The arithmetic is redone whenever the data changes.

5 Metrics and their statistical power

Full definitions are in the glossary; the reliability ordering is the substance. **chan%** is decision-grade within a channel set: seed spread ±0.5 points, so differences $\geq$ 2 points are real. The **fixed-holdout probe** r (36 held-out months, effective sample ≈ 15 after autocorrelation) has a 95% CI of ±0.57 — a compass, never a verdict; the same configuration has drawn 0.09 and 0.33 on two seeds. The **year-blocked k-fold** tests all 240 RAPID months once each (SE ≈ 0.10) and is the only probe number we argue from. The most instructive failure: a d_z=128 codec set the all-time fixed-holdout record (0.528) while the k-fold read 0.166 — the coarse instrument flatters widest exactly where overfitting is easiest.

6 Related work

Three literatures border this project (full survey: repository file `ml/LITERATURE.md`). *AMOC reconstruction from observables*: regression methods reach $r \approx 0.95$ against RAPID only after 18-month low-pass filtering and in-sample calibration [1, 2]; the machine-learning line trains on simulations and reports up to $r \approx 0.98$ inside a laminar state estimate [3] but drops to explained variance of 0.44–0.74 in eddy-resolving models [4, 5] and is essentially never applied to real RAPID data. Our $r = 0.60$ — monthly, deseasonalized, real data, blocked cross-validation, no RAPID-derived inputs — appears to be the first of its exact kind, and sits where the eddying simulations predict it should (the attention-head read-out reaches 0.69, with the attribution caveats of §7.3). *Forecast verification*: we adopt the decadal-prediction conventions [6] (MSSS, ACC, damped persistence). *Foundation models*: masked pre-training in anomaly space with frozen-backbone evaluation echoes Prithvi WxC and the frozen-decoder evaluation philosophy of [7]; ridge probes from frozen embeddings to integrated circulation indices, and missingness-as-information over the historical observing system, appear to be unoccupied territory.

7 Results

7.1 The bottleneck: what width buys

Fig. 1: the transport read-out wants more width than field reconstruction rewards, but not unboundedly — the turnover at d_z=128 is a truth-data limit, not a representation limit. That this sweep was run at C=12 becomes important in §7.4.

7.2 The probe ranking, and what the wind already knows

Fig. 2 is the project's headline and its conscience in one picture. The wind channels moved the probe more than every architecture decision combined (0.31 → 0.60); the global codec matches the regional one exactly (0.602 vs 0.604 — the generic-embedding hypothesis holds at fixed capacity); and the wind-only baseline at 0.53 says most monthly skill is Ekman physics. The first family's embedding increment over pure wind is +0.07; its distinctive contributions are the low-frequency component (18-month low-passed

run	window	C	patch	d_z	steps	chan%†	k-fold r [95% CI]	dip
first family — 4 Argo levels								
pilot4	NA	4	1	32	8k	+29.3	—	—
dz8	NA	12	1	8	30k	+28.6	0.111 [0.01, 0.20]	—
dz16	NA	12	1	16	30k	+30.3	0.151 [0.01, 0.28]	—
dz32	NA	12	1	32	30k	+30.5	0.182 [0.05, 0.31]	—
dz64	NA	12	1	64	30k	+30.6	0.308 [0.13, 0.46]	16%
dz128	NA	12	1	128	30k	+30.2	0.166 [0.07, 0.30]	—
wind14	NA	14	1	64	30k	+35.6	**0.604** [0.47, 0.72]	50%
global14	global	14	1	64	30k	+30.6	**0.602** [0.46, 0.73]	50%
global14b*	global	14	1	64	30k	+30.9	0.556 [0.43, 0.68]	—
global15sst	global	15	1	64	30k	+30.8	0.582 [0.43, 0.71]	47%
second family — 8 Argo levels (§3.1)								
patch24	global	24	**3**	64	40k	+32.0‡	0.543 [0.43, 0.66]	27%
pixel25	global	25	1	64	40k	+30.9‡	0.536 [0.38, 0.68]	**59%**
pixel24	global	24	1	64	40k	—	0.515 [0.41, 0.63]	—
pixel24-30k	global	24	1	64	30k	—	0.585 [0.45, 0.71]	—
pixel24-60k	global	24	1	64	60k	—	0.549 [0.42, 0.67]	—
patch24-30k	global	24	3	64	30k	—	0.551 [0.42, 0.68]	—
patch24-1M	global	24	3	64	**1M**	+32.1‡	0.571 [0.42, 0.70]	—
big10M (2 runs)	global	24	3	64	30k/60k	*training at time of writing*		

Table 2: **Master table: every codec run.** Generated from each run's own JSON by `ml/make_table.py`, not transcribed. †chan% is comparable only within one channel set (its persistence baseline moves with the channels); cross-run comparison goes through the year-blocked k-fold RAPID correlation. Dip = share of the winter 2009–10 collapse amplitude captured out-of-fold. The wind-stress-only ridge baseline (no embedding) scores $r = 0.531$ — the line every codec must beat. k-fold r here is the POOLED RIDGE rung of the probe ladder (the comparable-across-time instrument); the unpooled attention-head values (§7.3) are higher for every codec measured: global14 0.635, pixel25 0.617, patch24 **0.690**. *global14b is an exact-configuration replication on a different runner: it measures codec-training seed noise on the k-fold at $\approx \pm 0.05$ — the four pixel24/patch24 rows, spanning 0.515–0.585 across 30k–60k steps, are consistent with that noise and show no step-count trend; the patch24-1M row settles the saturation question directly — every probe metric is flat from the first 50k-step probe to the millionth step (§9), so the pilot size is capacity-limited, not compute-limited. ‡in-training probe at the final step, not the post-hoc trainprobe used for first-family rows. Stage-2 figures for global14b and global15sst are *withdrawn*, not stale: both were trained on a poisoned embedding cache (§9).

$r = 0.64$) and event anatomy (Fig. 7). In Sverdrups: RMSE 2.23 Sv against a target standard deviation of 2.79 Sv.

7.3 The probe ladder: pooling was hiding the signal

Fig. 3 is this revision's central result, and it changes both a number and a story. The number: the patch codec read by the attention head reaches $r = 0.690$ [0.570, 0.781], RMSE 2.02 Sv — against 0.60 for the best pooled-linear configuration, and against a wind-only baseline of 0.531 that the pooled second-family probes had barely cleared. The story: the ladder's two rungs dissociate cleanly. Pointwise nonlinearity (MLP rung) recovers nothing on any target for any codec — so the ridge was never the limit as a *function class*. Spatial structure (head rung) recovers a great deal — so the limit was the *pooling*. Mean-pooling the section computes the one statistic in which an east-west difference cancels, and thermal wind says the transport *is* an east-west difference. The two architectural changes of this cycle turn out to be halves of one physical idea: the 3×3 patch puts local gradients into each embedding, the attention head reads gradients across the section, and each looked worthless without the other — the patch codec's pooled numbers were indistinguishable from its pixel control.

The attribution matrix. An end-to-end baseline completes the picture: the *same* attention head, same folds, same regularization, fed raw section anomalies (values + observed-flags) instead of embeddings, at both receptive fields:

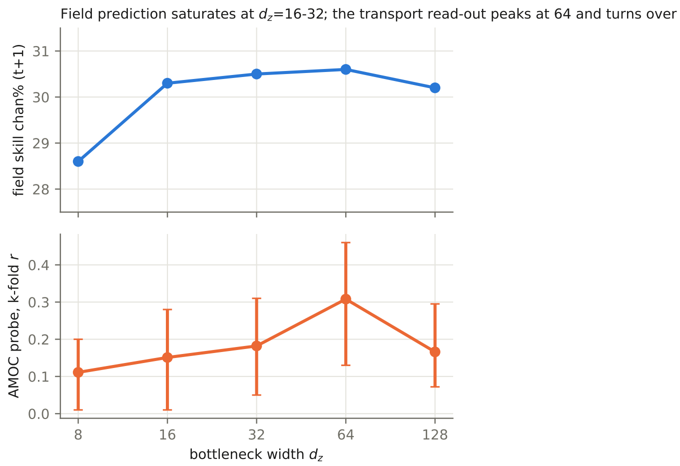

Figure 1: Bottleneck width d_z versus its two customers, on the 12-channel North-Atlantic tensor at matched training steps. Field prediction (top) saturates by d_z=16–32; the AMOC probe (bottom, with 95% block-bootstrap CIs) rises to a peak at d_z=64 and turns over at 128 — 128 ridge features against ∼220 truth months per fold overfits. d_z=64 became the working setting, a choice made when the tensor had 12 channels.

tokens carry	raw data	codec embedding
single pixel	0.613 [0.48, 0.73]	0.617 [0.49, 0.73]
3 × 3 neighbourhood	0.659 [0.55, 0.75]	**0.690** [0.57, 0.78]

The headline decomposes, in descending order: *unpooling* the section is the largest term (pooled ridge 0.54 → head 0.61+); the 3 × 3 *receptive field* adds ∼+0.045 and does so on raw data alone; *pretraining* adds +0.031 at matched receptive field — positive in direction, not statistically significant — and nothing at a single pixel. For the monthly RAPID probe, a supervised head over raw data recovers nearly everything. The pretraining case therefore rests on the scoreboards where 240 labelled months have no answer: field prediction (stage 2 beats persistence everywhere; the matching end-to-end field-forecaster baseline is the open follow-up), transfer to sparsely-labelled targets like MOVE, and the open question of whether the pretraining margin grows with codec capacity — the 10M-parameter codecs now training will say.

Honesty box: the head is the most expressive probe we run (∼25k parameters against ∼220 training months per fold), and although it is regularized hard, early-stopped on a train tail, and seed-averaged behind the same blocked folds as every other number, the head-vs-ridge gaps have not yet been given a paired significance test, and the 0.690 awaits replication on a second codec seed; both are queued. A capacity sweep of the head itself (23k→325k→2.0M parameters) shows *neither* side is parameter-limited: at 325k both columns drop (raw-3 × 3 0.659 → 0.590, embedding-3 × 3 0.690 → 0.611), as the label-count arithmetic predicts (∼220 training months per fold support a head of tens of thousands of parameters, not hundreds), and the embedding margin persists at both sizes. The 2.0M cells are running as a formality.

7.4 Where more data moved the skill

Under *pooled* probes, expanding the input appeared to cost RAPID skill: both second-family codecs landed on the wind-only line (0.543 and 0.536 against 0.531) where the first family held +0.07. Section 7.3 resolves this: the skill had moved into cross-pixel structure, not out of the representation, and the unpooled read-out recovers it in full (0.617–0.690). The RAPID half of this subsection's original claim is therefore retracted; what follows — the MOVE result — is the part that survives, and it was measured pooled, so unpooling can only strengthen it.

At MOVE the same channels do the opposite. The 25-channel codec scores $r = 0.206$ [0.044, 0.346] — the first multi-target result in this programme whose confidence interval excludes zero — with an 18-month low-passed correlation of 0.623, against 0.111 and 0.379 for the 15-channel codec. MOVE is

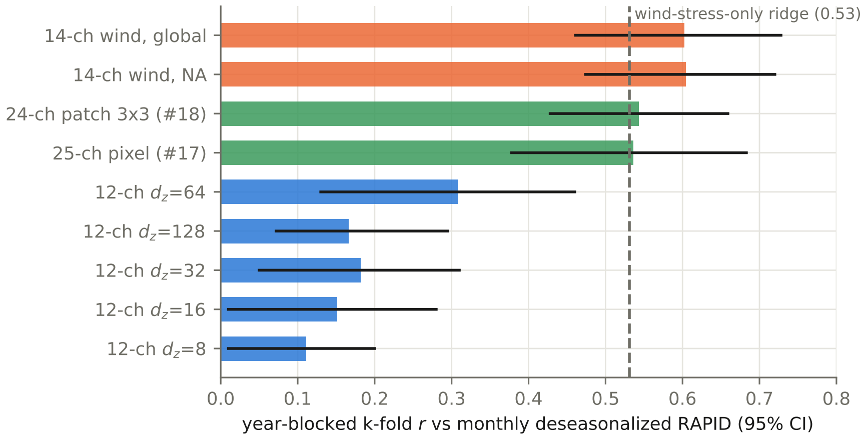

Figure 2: Year-blocked k-fold correlation with monthly deseasonalized RAPID transport, all codec generations, 95% CIs. Adding wind stress (orange) nearly doubled the defensible probe; training globally cost the Atlantic read-out nothing. The second-family codecs (green) fall back onto the dashed wind-only baseline — a ridge on section-mean wind stress with no embedding at all.

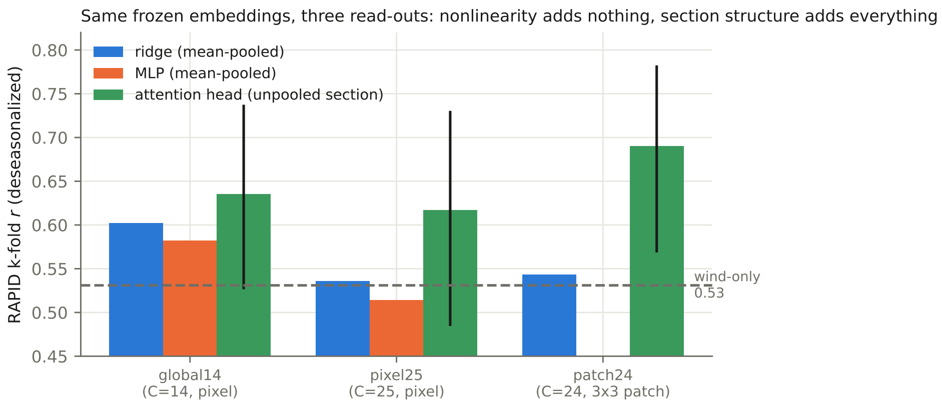

Figure 3: Same frozen embeddings, three read-outs, same year-blocked folds. The MLP (orange) trails the ridge (blue) everywhere — pointwise nonlinearity recovers nothing. The attention head over the unpooled section (green, with 95% CIs) beats every pooled probe, restores the 25-channel codec to parity with the 14-channel one, and reveals the patch codec — invisible to pooled probes — as the strongest representation in the programme at $r = 0.690$.

a deep, density-driven transport; RAPID's monthly variance is largely Ekman. The added channels are Argo T and S to 1900 dbar. The skill went where the physics says the information is.

The corrected reading is about read-outs, not bottlenecks: the embedding kept everything, and the probe determined what could be seen. For the generic-embedding hypothesis of §6 this is a *stronger* confirmation than the regional-versus-global comparison — one task-blind representation now serves the wind-dominated and the density-dominated target simultaneously, provided the read-out respects the structure of the question being asked of it.

One more asymmetry is worth recording: the 25-channel codec captures **59%** of the winter 2009–10 collapse out-of-fold, the best of any run in the programme, while its monthly correlation is the second worst. Event anatomy and average correlation are not the same capability, and a table that reports only the latter will hide the former.

7.5 How far ahead the ocean can be predicted

Fig. 5: feeding predictions back month after month, the anomaly field stays predictable beyond a year (MSSS +0.11 at h=12; ACC 0.36). The decay is decorrelation, not amplitude damping (the forecast-

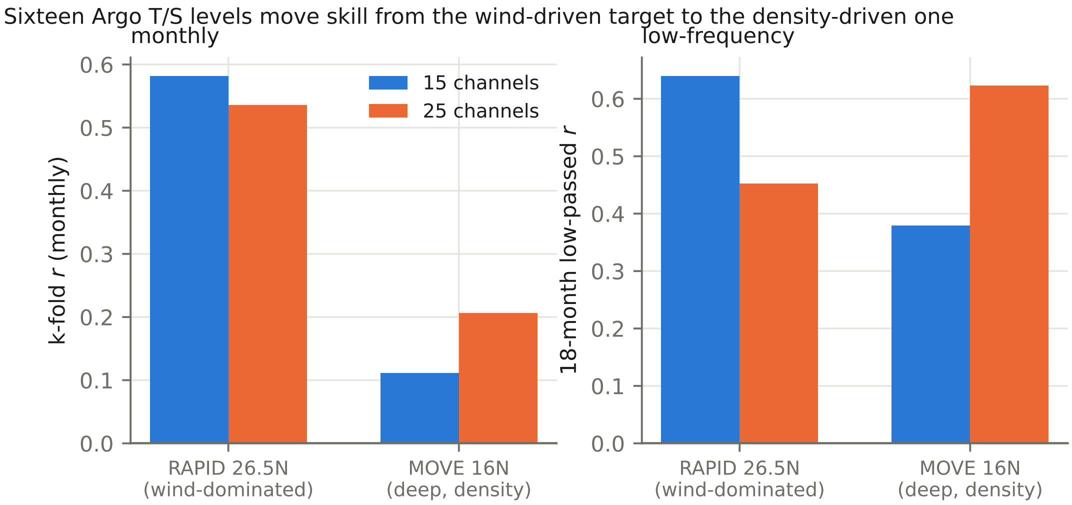

Figure 4: Sixteen Argo temperature and salinity levels to 1900 dbar *appeared* to cost monthly RAPID skill under pooled probes (left, first pair) — resolved by §7.3 as a read-out artifact, not a representation change — while genuinely roughly doubling skill at MOVE (left, second pair) and transforming the low-frequency read-out there (right). MOVE measures the *deep* western-boundary flow at 16°N — precisely the quantity the added levels describe.

to-observed amplitude ratio holds near 0.5 throughout). The AMOC read-out on rolled embeddings, by contrast, dies after about a season — transport forecasting at range remains open, consistent with its wind-driven monthly variance.

7.6 Stage-2 capacity and the training curves

7.7 Case study: the winter 2009–10 collapse

8 The static-channel lesson

An ablation found the two climatology input channels contribute *exactly nothing* to any metric: a static map is a function of position, the encoder receives position, and so the channel carries no information the model does not already have. The general rule — a channel is worth what position and season cannot explain of it — now gates every data-acquisition decision. Climatological knowledge is still essential, but it enters as preprocessing arithmetic (the anomaly transform), never as a feature the network must carry through its bottleneck.

9 Limitations and open questions

(1) The wind-only baseline bounds honest claims: monthly AMOC skill is substantially Ekman, though the unpooled head now clears the baseline by +0.10–0.16 where the pooled probes had shrunk the margin to noise. (2) The head result is new and strong, and strength is when scrutiny matters most: no paired significance test yet, no second codec seed yet (a replicate is queued on the fleet), and the head has been run only on RAPID — the MOVE and low-frequency versions may move too. (3) d_z=64 was chosen on a 12-channel tensor and carried unchanged into 24–25 channels; the sweep has not been repeated at the new width. (4) Absolute k-fold values are mildly optimistic — the codec saw non-holdout months' *fields* during self-supervised pretraining (transports were never seen by anything); codec-to-codec comparisons are unaffected. (5) The rollout's horizon is measured only to the holdout-year boundary; the crossover to no skill lies beyond 12 months and is not yet located. (6) Transport forecasting at range is unsolved. (7) *A methodological casualty worth recording*: the stage-2 results for two runs were withdrawn after we found their embedding cache had been populated by a different codec — the shape check (T, P, d_z) matched, so training proceeded happily on the wrong representation, and

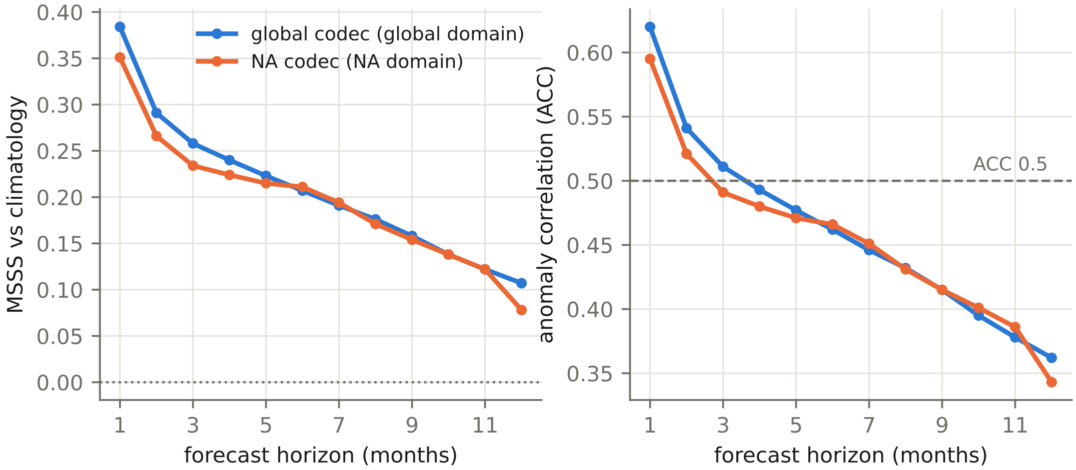

Figure 5: Autoregressive rollout skill versus forecast horizon for the global codec (blue, global domain) and the regional codec (orange, North-Atlantic domain; domains differ, so compare shapes more than levels). Left: MSSS against climatology — still positive at 12 months. Right: anomaly correlation, crossing the conventional 0.5 useful-skill line near four months.

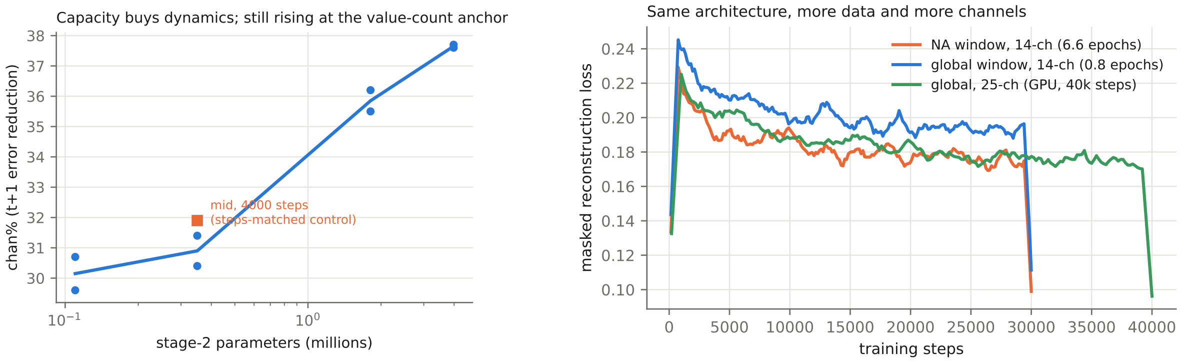

Figure 6: Left: stage-2 capacity sweep (both seeds shown per size) — width×depth buys real dynamics, still rising at 4M parameters; the steps-matched control (orange square) separates capacity from training length. Right: codec training curves across generations.

the signature was healthy embedding-space skill beside catastrophic decoded skill. The cache key now carries a hash of the codec's weights, so a stale cache is a miss rather than a lie. We report the affected numbers as withdrawn rather than regenerating them, because both codecs belong to the superseded first family.

The training-time ceiling has now been located directly: a 1,000,000-step run at pilot size (LR decayed to a constant 10% floor by 50k steps, probes every 50k) completed in 2.6 h on one RTX 4090 and is *flat from the first probe to the last* — linear r oscillates in $[0.43, 0.48]$, channel-space skill vs. persistence is pinned at 32%, and the final checkpoint's k-fold (0.571) sits within seed noise of the 40k-step run. Twenty-five times the compute of the pilot regime buys nothing at 0.92M parameters: the pilot is capacity-limited, which is what the Chinchilla arithmetic predicted (∼13M parameters for this tensor) and what the 10M-parameter codecs now training are for. Next levers, in expected-value order: paired significance and seed replication for the head results; the head probe at MOVE and at low frequency; re-running the d_z sweep at C=24; then the approved data ladder — (a) extending the time axis to 1982, where monthly SST and wind exist pre-GLORYS and the Florida cable's 1982–92 decade becomes truth, (b) 0.25° resolution for the GLORYS-derived channels (16× the pixels), (c) doubling the Argo pressure levels — with the Chinchilla arithmetic redone and the codec re-sized at each rung; and label-efficiency curves over the 792 non-RAPID truth months.

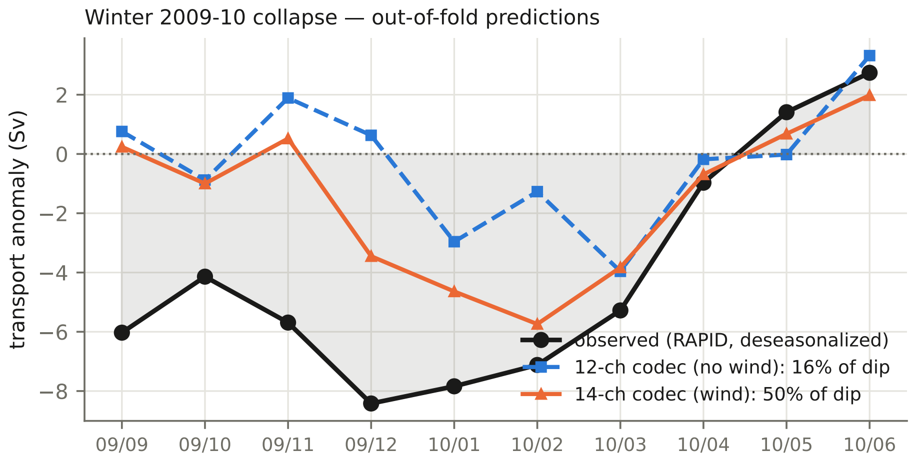

Figure 7: The sharpest event in the RAPID record, predicted out-of-fold. Without wind channels (blue) the model captures 16% of the dip amplitude; with them (orange), 50%. The 25-channel codec reaches 59% (not plotted; see Table 2). The November onset — triggered inside a month — stays invisible to monthly data.

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Reproduction: every number in this report is generated by scripts in `ml/` of github.com/blauewelt/earth (see `ml/FINDINGS.md` §8); the master table by `ml/make_table.py`; figures by `ml/paper/make_figs.py`. Raw source data is mirrored in the repository's `data-cache-v1` release (RG Argo, cite Roemmich & Gilson 2009; NCEP R1, public domain); a DOI-carrying archive (Zenodo) is planned once the in-flight results settle.